Lulu Zhang

Doctor of Philosophy (PhD) in Astrophysics

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Research Interests and Goals

My research focuses on spatially resolved studies of PAH, ionized-gas, and H_2 emission features using JWST spectroscopy to probe physical conditions across diverse galactic environments. I develop and apply multi-phase diagnostics that combine observational measurements with theoretical predictions to investigate star formation, AGN activity, and their associated feedback mechanisms, particularly shocks and outflows. Ultimately, I aim to deepen our understanding of the physical mechanisms underlying AGN feedback and its role in shaping galaxy evolution. I am also exploring how AI tools can accelerate this research and contributing to preparatory efforts for next-generation flagship observatories (i.e., HWO and TMT).

Education and Professional Appointments

Research Assistant Professor **2026 Sep –**
UT San Antonio, San Antonio, USA

Postdoctoral Researcher **2023 – 2026**
UT San Antonio, San Antonio, USA
Faculty Host: Prof. Chris Packham

Ph.D., Astrophysics **2017 – 2023**
Peking University, Beijing, China
Dissertation: *The Spatially Resolved Mid-infrared Spectral Diagnostics in Nearby Galaxies*
Advisor: Prof. Luis C. Ho

B.S., Applied Physics **2013 – 2017**
Shandong University, Weihai, China

Publication Summary

33 papers in total (**10** first-author) with over **550** citations (see SciX by ADS; as of November 2026)
h-index: **14** (SciX; as of November 2026)

Approved Proposals

ISSI International Team Call 2026 (26-681; invited member) **2026**
PAHs Across Cosmic Environments: JWST and Galaxy Evolution *Duration: 2026-2028*

JWST Cycle 4 (GO-07429; P.I.) **2025**
Unraveling Effects of Different Modes of AGN Feedback via PAH Features *Allocation: 11.7 hours*

JWST Cycle 4 (GO-07802; Co-I.) **2025**
Dust in shocks: The missing link in AGN feedback *Allocation: 14.2 hours*

JWST Cycle 4 (GO-07195; Co-I.) **2025**
Deciphering the torus and extended dust properties of local active galactic nuclei *Allocation: 32.8 hours*

JWST Cycle 3 (GO-04972; P.I.) **2024**
Hunting the Kinetic Mode Feedback of AGNs via PAH Features *Allocation: 12.4 hours*

JWST Cycle 3 (GO-05017; Co-I.) **2024**
AGN and their outflows: probing fragmentation and survival of PAH molecules *Allocation: 18.7 hours*

JWST Cycle 2 (DD-06659; Co-I.) **2024**
Probing Cold Dust Survival, Hot Dust Formation, and Mass-Loss of Type Ibn SN *Allocation: 6.3 hours*

Research Funding

NASA Shared Services Center (80NSSC27K0018; co-I.) **2026 – 2028**
AGN and Deconvolution for HWO *Amount: $ 395.7 k*

Space Telescope Science Institute (JWST-GO-07429; P.I.) **2025 – 2028**
Unraveling Effects of Different Modes of AGN Feedback via PAH Features *Amount: $ 120.2 k*

Space Telescope Science Institute (JWST-GO-04972; P.I.) **2024 – 2027**
Hunting the Kinetic Mode Feedback of AGNs via PAH Features *Amount: $ 129.7 k*

Selected Awards

2024– Postdoctoral Research Fellow of the Year, UTSA
2017–23 Award for Outstanding Doctoral Graduates of PKU
Award for Outstanding Doctoral Dissertation of PKU (Advisor: Prof. Luis C. Ho)
2013–17 Award for Outstanding Undergraduates of SDU, Weihai
Award for Outstanding Undergraduates of Shandong Province
Award for Outstanding Bachelor's Thesis of SDU (Advisor: Prof. Lulu Fan)

Service and Outreach

2026 Invited Judge for "UTSA Postdoctoral Research Fellow of the Year Award"
2026–28 Invited International Space Science Institute International Team Member
2025&26 Invited NASA/ROSES Review Panelist
2024 Invited Judge for "UTSA Final 3MT Research Competition"
2024–25 HWO AGN Work Group Steering Committee Member (sub-group leader)
2023– TMT International Science Development Team Member
2023– GATOS Collaboration Associate then Full Member
2022– Referee for AJ, ApJ, ApJL, A&A, and MNRAS
2019–22 Cashier for Graduate Fund (Donated by Graduates) of KIAA
2019–20 Admin for Astro-wiki Page of Biweekly ISM&Galaxy Group Meeting at KIAA
2019 LOC for International Conference – "Cosmic Evolution of Quasars"
2018–19 Organizer for Weekly "Happy Hour" of KIAA
2018&19 Secretary for Graduate Student Interview of KIAA

Teaching and Mentoring

Part-Time Lecturer **Fall 2026**
AST3013: Fundamentals of Astronomy (Undergraduate Course) *UT San Antonio*

- 1.25-hour lecture on "Astronomical Detection of Light"
- 1.25-hour lecture on "Solar System in Perspective"
- 1.25-hour lecture on "Properties of Stars"

Part-Time Lecturer **Spring 2026**
AST1013: Introduction to Astronomy (Undergraduate Course) *UT San Antonio*

- 2.5 hours of lectures on "Jovian Planet Systems"
- 2.5 hours of lectures on "Lives of Stars"

Teaching Assistant **Spring 2020**
Astrophysics of Diffuse Interstellar Medium (Graduate Course) *Peking University*

⋆Students Advised

John Schneider, *graduate student at UT San Antonio, as Ph.D. committee member* **2024–**
William Olson, *undergraduate student at UT San Antonio, working on AI-related AGN projects* **2024–**

Seminars, Conferences, & Workshops

2026 July "Young Astronomers Learning LSST (YALL) workshop", College Station, USA
2026 May "Torus Workshop 2026: Multi-wavelength Approach to The AGN Structures", Kyoto, Japan (oral talk)
2026 Apr "COS Symposium, UTSA", San Antonio, USA (oral talk)
2026 Apr "Physics and Astronomy Seminar, Mizzou", Online (virtual talk; invited)
2026 Mar "Prompt by Design: Processing Real-World Multimodal Data with AI", San Antonio, USA
2026 Jan "Physics and Astronomy Seminar, UTSA", San Antonio, USA (oral talk)
2026 Jan "Inaugural Session of NASA AGN Science Interest Group Spotlight Series", Online (virtual talk; invited)
2025 Nov "The Fifth Pan Dust Conference", Tucson, USA (poster talk)
2025 Oct "Dust and Gas in the HST and JWST Era", Toruń (hybrid), Poland (virtual talk; invited)
2025 Oct "Infrared Spectroscopy from Space: New Frontiers from Exoplanets to the Early Universe", Pasadena, USA (oral talk)

2025 June "AAS 246th Annual Meeting", Anchorage, USA (oral talk & judge of poster section)
2025 May "GATOS Meeting - Tenerife 2025", Tenerife, Spain (oral talk)
2024 Nov "HWO AGN Working Group Presentation", Online (virtual talk; invited)
2024 Jan "Extra Galaxy Evolution Seminar, Oxford", Oxford, UK (oral talk)
2024 Jan "GATOS Workshop - Madrid 2024", Madrid, Spain (oral talk)
2024 Jan "GATOS Meeting - Madrid 2024", Madrid, Spain (oral talk)
2023 Nov "Physics and Astronomy Seminar, UTSA", San Antonio, USA (oral talk)
2023 Sep "The First Year of JWST Science Conference", Baltimore, USA (poster talk)
2023 May "The 25th Guoshoujing Workshop and the Galaxy & Cosmology Frontier Workshop", Huangshan, China (oral talk, session chair)
2021 Dec "Chinese Astronomical Society Annual Meeting", Nanchong, China (oral talk)
2021 Nov "KIAA Forum on Gas in Galaxies: for Early Career Scientists", Online (oral talk)
2021 Oct "East Asia AGN Workshop 2021", Chongqing, China (poster)
2021 June "The 11th C-MaNGA Workshop and the CSST-IFS Science Meeting", Zhuhai (postponed), China (oral talk, session chair)
2020 Oct "Supermassive Black Hole Workshop", Lijiang, China (oral talk)
2019 Sep "KIAA Forum on Gas in Galaxies: Probing the Activities and Power Engines from Local to Distant Universe", Beijing, China (poster)
2019 Sep "Chinese Astronomical Society Annual Meeting", Delingha, China (oral talk)
2019 Jan "East Asia AGN Workshop 2019", Taipei, China (poster)
2018 July "Thirty Meter Telescope-MICHI and China: a Workshop on Thermal-IR Astronomy", Beijing, China (oral talk)
2018 July "The 20th Guoshoujing Workshop and the Galaxy & Cosmology Frontier Workshop 2018", Sanmenxia, China (poster)

Publication List

⋆ **As first-author:**

[10] **Zhang, L.**, Packham, C., Hicks, K. S. E., et al., "GATOS: Distinct Feedback Modes in AGN Central Regions Revealed by Spatially Resolved JWST Spectroscopy", 2026, ApJL, 1007, L52
[9] **Zhang, L.**, Packham, C., Hicks, K. S. E., et al., "Evidence of Feedback Effects in Low-luminosity Active Galactic Nuclei Revealed by JWST Spectroscopy", 2026, ApJL, 998, L32
[8] **Zhang, L.**, Davies, I. R., Packham, C., et al., "Theoretical Diagnostics for the Physical Conditions in Active Galactic Nuclei under the View of JWST", 2025, ApJS, 280, 65
[7] **Zhang, L.**, García-Bernete, I., Packham, C., et al., "Polycyclic Aromatic Hydrocarbon Emission in the Central Regions of Three Seyferts and the Implication for Underlying Feedback Mechanisms", 2024, ApJL, 975, L2
[6] **Zhang, L.**, Packham, C., Hicks, K. S. E., et al., "The Galaxy Activity, Torus, and Outflow Survey (GATOS). IV. Exploring Ionized Gas Outflows in Central Kiloparsec Regions of GATOS Seyferts", 2024, ApJ, 974, 195
[5] **Zhang, L.**, & Ho, L. C., "The Interaction between AGN and Starburst Activity in the Circumnuclear Region of NGC 7469 as Viewed with JWST", 2023, ApJL, 953, L9
[4] **Zhang, L.**, & Ho, L. C., "Estimating Molecular Gas Content in Galaxies from Polycyclic Aromatic Hydrocarbon Emission", 2023, ApJ, 943, 1
[3] **Zhang, L.**, & Ho, L. C., "The Contribution of Evolved Stars to Polycyclic Aromatic Hydrocarbon Heating and Implications for Estimating Star Formation Rates", 2023, ApJ, 943, 60
[2] **Zhang, L.**, Ho, L. C., & Li, A., "Evidence that Shocks Destroy Small PAH Molecules in Low-luminosity Active Galactic Nuclei", 2022, ApJ, 939, 22
[1] **Zhang, L.**, Ho, L. C., & Xie, Y., "A Method to Extract Spatially Resolved Polycyclic Aromatic Hydrocarbon Emission from Spitzer Spectra: Application to M51", 2021, AJ, 161, 29

† **As significant contributing-author:**

[3] Schneider, J., Packham, C., **Zhang, L.**, et al., "Torus Models of NGC 4388: an Elevated MIR/submm Emission Ratio", 2026, ApJ, submitted
[2] Delaney, E. D., Hicks, K. S. E., **Zhang, L.**, et al., "GATOS XIII. Outflow Driven Excitation in Seyferts", 2026, ApJ, 1002, 20
[1] Delaney, E. D., Hicks, K. S. E., **Zhang, L.**, et al., "Excitation of Molecular Hydrogen in Seyferts: NGC 5506 and NGC 3081", 2025, ApJ, 993, 217

‡ **As contributing co-author:**

[20] Veenema, O., Rigopoulou, D., Thatte, N., et al., "Beyond the S-lines – New JWST H2 O-line Excitation Diagnostics Point to X-ray Heating in the Active Galaxy NGC 3227", 2026, MNRAS, submitted

[19] Rosario, D., Haidar, H., Campbell, S., et al., "GATOS N: Extended circumnuclear dust emission in nearby Seyfert galaxies surveyed by JWST/MIRI", 2026, MNRAS, 552, 1

[18] Donnan, F. R., García-Bernete, I., Rigopoulou, D., et al., "GATOS XIV: The first direct kinematic evidence of dusty outflows from AGN via PAH kinematics of local Seyfert galaxies with JWST", 2026, MNRAS, 550, 1

[17] Veenema, O., Thatte, N., Rigopoulou, D., et al., "Decoupling the AGN outflow and star-forming disk kinematics in the nuclearregion of NGC 7582 with JWST NIRSpec and MIRI/MRS", 2026, MNRAS, 548, 1

[16] Hermosa Muñoz, L., González Fernández, J. R., Alonso-Herrero, A., et al., "Unveiling physical processes in local active galaxies: Unsupervised hierarchical clustering of JWST MIRI/MRS observations", 2026, A&A, 708, 297

[15] Haidar, H., Rosario, D., García-Bernete, I., et al., "GATOS XI. Excess dust heating in the Narrow Line Regions of nearby AGN revealed with JWST/MIRI", 2026, MNRAS, 546, 1

[14] Riffel, R. A., Colina, L., Costa-Souza, J. H., et al., "Impact of AGN and nuclear star formation on the ISM turbulence of galaxies: Insights from JWST/MIRI spectroscopy", 2026, A&A, 705, 59

[13] Veenema, O., Thatte, N., Rigopoulou, D., et al., "Shock-driven heating in the circumnuclear star-forming regions of NGC 7582: Insights from JWST NIRSpec and MIRI/MRS spectroscopy", 2025, MNRAS, 544, 3361

[12] Esposito, F., Alonso-Herrero, A., García-Burillo, S., et al., "GATOS X. Molecular gas clumpiness under the influence of AGN", 2026, A&A, 705, 33

[11] Campbell, S., Rosario, D., Haidar, H., et al., "GATOS IX. A Detailed Assessment and Treatment of Emission Line Contamination in JWST/MIRI Images of Nearby Seyfert Galaxies", 2025, MNRAS, 544, 648

[10] Lopez-Rodriguez, E., Ramos Almeida, C., Pereira-Santaella, M., et al., "GATOS VIII. On the physical origin of the extended MIR emission in AGN", 2025, ApJ, 994, 206

[9] González-Martín, O., Díaz-González, J. D., Martínez-Paredes, M., et al., "JWST reveals the diversity of nuclear obscuring dust in nearby AGN: nuclear isolation of MIRI/MRS data cubes and continuum spectral fitting", 2025, MNRAS, 539, 2158

[8] Esparza-Arredondo, D., Ramos Almeida, C., Audibert, A., et al., "Molecular gas stratification and disturbed kinematics in the Seyfert galaxy MCG-05-23-16 revealed by JWST and ALMA", 2025, A&A, 693, 174

[7] Fuller, L., Lopez-Rodriguez, E., García-Bernete, I., et al., "GATOS VII. The 20-214 μm imaging atlas of active galactic nuclei using SOFIA", 2025, ApJS, 276, 64

[6] García-Bernete, I., Rigopoulou, D., Donnan, F. R., et al., "GATOS V. Unveiling PAH survival and resilience in the circumnuclear regions of AGN with JWST", 2024, A&A, 691, 162

[5] Hermosa Muñoz, L., Alonso-Herrero, A., Pereira-Santaella, M., et al., "A biconical ionised gas outflow and evidence for positive feedback in NGC 7172 uncovered by MIRI/JWST", 2024, A&A, 690, 350

[4] García-Burillo, S., Hicks, K. S. E., Alonso-Herrero, A., et al., "Deciphering the imprint of AGN feedback in Seyfert galaxies: Nuclear-scale molecular gas deficits", 2024, A&A, 689, 347

[3] Haidar, H., Rosario, D., Alonso-Herrero, A., et al., "Dust beyond the torus: Revealing the mid-infrared heart of local Seyfert ESO 428-G14 with JWST/MIRI", 2024, MNRAS, 532, 4645

[2] Davies, R., Shimizu, T., Pereira-Santaella, M., et al., "GATOS: missing molecular gas in the outflow of NGC 5728 revealed by JWST", 2024, A&A, 689, 263

[1] Leist, M., Packham, C., Rosario, D., et al., "Deconvolution of JWST/MIRI Images: Applications to an Active Galactic Nucleus Model and GATOS Observations of NGC 5728", 2024, AJ, 167, 96

⋆ **Other publications:**

[3] Dressing, C. D., et al., "Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory", 2026, arXiv:2608.11294

[2] **Zhang, L.**, et al., "Deciphering The Launching of Multi-phase AGN-driven Outflows and Their (Spatially Resolved) Multi-scale Impact", 2026, Astronomical Society of the Pacific Conference Series, 542, 161; arXiv:2508.01437

[1] Skidmore, W., et al., "Thirty Meter Telescope International Observatory Detailed Science Case 2024", 2024, arXiv:1505.01195v3

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